

HW5 GPS LEAST SQUARES POINT SOLUTION

Assigned: 5 October 2026

Due: 16 October 2026

OBJECTIVE

The objective of this assignment is to compute a single GPS least squares position solution and investigate its accuracy, convergence from a poor initial guess, and the effect of a low-elevation satellite.

DATA

- Date: Wednesday, August 19, 2026 = GPS Week 2432 (or 384), TOW 259200-345600/ DOY 231
- User location: NIST (in Boulder, CO): ECEF coordinates: [-1288398.567 -4721696.932 4078625.350] (m)
 - Find the transformation matrix from ECEF to ENU using this location as the reference point.
- RINEX File: NIST00USA_R_20262310000_01D_30S_MO.rnx
- Broadcast ephemeris file: brdc2310.26n (DOY is the 3-digit day of year)

ASSIGNMENT

1. Gather data for a single point solution.

- Start with the correct location for NIST given above as your *a priori* position ($\vec{r}_0$).
- For the first time (epoch) in the NIST observation file, gather and compute the following values for each GPS satellite tracked and fill in the table below:
 - ionosphere free pseudorange observable PIF (using C1C and C2W) in meters (cm precision)
 - expected range based on the *a priori* location in meters (cm precision)
 - elevation and azimuth (based on adjusted satellite location at T_i) in degrees (0.1 deg precision)
 - satellite clock correction (not including relativity) in meters (cm precision)
 - relativity correction (from the orbital elements or r, v of the GPS satellite) in meters (cm precision)
 - tropospheric model (used in HW4) in meters (cm precision)

PRN	PIF (m)	R1 (m)	EL (deg)	AZ (deg)	BSV (m)	RELAT (m)	TROP (m)
3							
4							
7							
8							
9							
16							
26							
27							
31							

2. Calculate the least squares point solution for this single epoch.

- Compute and print the prefit residuals ($\delta\rho$) for each satellite based on the data in your table above.
- Construct and print the measurement sensitivity matrix (called A or G) using the ECEF coordinates of the GPS satellites and the observer as discussed in lecture or in your book.
- Calculate and print out the least squares solution for the state vector corrections: $[\delta x \ \delta y \ \delta z \ \delta b]^T$. (Since you started with the right answer, this represents the best possible solution error.)
- Compute and display the post-fit pseudorange residuals: $\delta\rho_{post} = \delta\rho - G [\delta x \ \delta y \ \delta z \ \delta b]^T$ and discuss what you see.
- See if you can get a more accurate solution by removing low elevation satellites or any satellite with an unusually large post-fit residual. Describe your results.

3. Repeat the procedure you put together in Questions 1 & 2, but with an incorrect starting guess and iterate. Be sure to recompute the G matrix for based on the new position estimate at each iteration

- Start with an initial guess (a priori) for location of $[40^\circ, -105^\circ, 1600 \text{ m}]$. Compute the ECEF coordinates of this *a priori* position ($\vec{r}_0$) and the error vector ($\Delta\vec{r}_0$) from $\vec{r}_0$ to the true NIST location given above: $\Delta\vec{r}_0 = \vec{r}_{NIST} - \vec{r}_0$. What is the (straight line) distance between your guess and the true location?
- Gather the information as described in Question 1b using this *a priori* position and do the computations described in Question 2 to find the state vector corrections.
- Update your estimate of the position and compute the distance of this new position estimate from the true location of NIST. It should be smaller than your first guess.
- Iterate your solution (recomputing the values in the table, prefit residuals, and G matrix as necessary) until the magnitude of your state vector corrections is below 1 cm. Report the magnitude of the state vector correction and the updated position estimate for each iteration with a precision of 1 cm.
- What is your final estimate of the clock bias? Report the error in your final estimated position in ECEF coordinates for this time as $\Delta\vec{r}_f = \vec{r}_{NIST} - \vec{r}_f$. Compute and display the post-fit pseudorange residuals: $\delta\rho_{post} = \delta\rho - G [\delta x \ \delta y \ \delta z \ \delta b]^T$ and discuss what you see.
- Compute and report the error $\Delta\vec{r}_f$ in the ENU frame for NIST.
- Compute the HDOP and VDOP for the satellite geometry for this epoch.

4. Try initializing the a priori position at a location much farther away from the truth and iterate until it converges. Do you get the same final result? Explain why. (*Hint: Be careful with eliminating large outliers when starting from far away.*)